



# Computational Approach for Rapidly Predicting Temperature-Dependent Polymer Solubilities Using Molecular-Scale Models

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One promising approach to recycle multicomponent plastic waste (e.g., multilayer plastic films) is selective dissolution. Selective dissolution is a solvent-mediated process in which differences in polymer solubility in a carefully chosen solvent system are exploited to recover a target polymer. Here, a computational approach was developed that rapidly predicts temperature-dependent polymer solubilities to guide the design of solvent systems for solvent-mediated polymer recycling. Polymer conformations were obtained from molecular dynamics simulations by modeling the polymer as a short

oligomer and then used as input to the conductor-like screening model for real solvents (COSMO-RS) for solubility predictions. Using polyethylene (PE) and ethylene vinyl alcohol (EVOH) as representative polymers, the effect of simulation parameters was systematically studied, and predicted solubilities were found to be in good agreement with experimental measurements. The applicability of the approach was demonstrated by identifying selective solvents for PE and EVOH dissolution from a library of 524 solvents.

## Introduction

Multilayer plastic films are widely used materials in the packaging industry<sup>[1]</sup> with global production exceeding 100 million tons each year.<sup>[2]</sup> Multilayer films are formed from several layers of common polymers, each of which is selected to contribute useful properties to the film. For example, polyethylene (PE) is chemically inert and serves as a good moisture barrier; ethylene vinyl alcohol (EVOH) is a good gas barrier but a poor moisture barrier; polyvinyl chloride (PVC) has good cling behavior and puncture resistance but is oxygen permeable; and polyethylene terephthalate (PET) is rigid and impact resistant but has poor alkali resistance.<sup>[3–5]</sup> The manufacture of multilayer films employs methods such as co-extrusion, layer-by-layer assembly, extrusion lamination, extrusion coating, adhesive lamination, and solvent casting to combine different polymers for various applications.<sup>[6,7]</sup> Unfortunately, the resulting materials are extremely challenging to recycle because of their complex compositions and the incompatibility of their different polymer components.<sup>[8]</sup> Current plastic recycling technologies based on mechanical or chemical methods are designed for single-

component materials and are unable to recycle multilayer films into new products.<sup>[9–11]</sup> As a result, large amounts of multilayer plastic films accumulate as solid waste, including post-industrial waste that is unutilized during industrial fabrication processes (up to 40% of all films<sup>[12]</sup>) and municipal post-consumer waste. Only 13% of plastic municipal solid waste in the U.S. has been recycled with the majority of these films being landfilled.<sup>[13]</sup> There is thus an urgent need for new technologies for recycling multilayer plastic films.

Solvent-based polymer recycling processes are promising, near-term approaches that exploit differences in polymer solubility to recover polymer resins in pure form from carefully chosen solvent systems that dissolve only target polymers. A few solvent-based plastic recycling processes are being developed industrially. A German company, APK AG, developed a process called Newcycling to recycle low-density (LD)PE/polyaniline (PA) films.<sup>[14,15]</sup> A solvent-based PVC purification project, VinylLoop, was launched in Italy in 2002 but terminated in 2018 due to challenges in separating hazardous plasticizer additives.<sup>[16,17]</sup> Unilever and the Fraunhofer Institute launched a pilot project called CreaSolv that aims to recycle PE from packaging materials.<sup>[18,19]</sup> A Canadian startup called Polystyvert uses essential oils to recycle polystyrene (PS).<sup>[20]</sup> These examples demonstrate the industrial relevance and broad applicability of solvent-based polymer recycling technologies. However, limited information is available regarding the processing conditions and solvents used in these processes.

Recently, we developed the solvent-targeted recovery and precipitation (STRAP) process to recycle multiple polymer components from a multilayer plastic film.<sup>[8]</sup> The STRAP process selectively dissolves constituent polymers through a series of solvent washes, each designed to dissolve only a single polymer component. Pure resins are then precipitated from solution by changing the temperature and/or adding antisolvents.<sup>[21]</sup> We

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demonstrated that the STRAP process can successfully recover PE, EVOH, and PET resins from a post-industrial waste film using two solvent washes.<sup>[8]</sup> In the first solvent wash, toluene was used as a selective solvent and acetone as an antisolvent to recover PE. In the second solvent wash, dimethyl sulfoxide (DMSO) was used as a selective solvent and water as an antisolvent to recover EVOH. The remaining polymer component of the film was PET, which was recovered after removal of the other polymer components. The STRAP process highlights the feasibility of recycling multiple polymers from complex multilayer films.

There are several important design considerations underlying a solvent-based polymer recycling process like STRAP, including the selection of an appropriate solvent and antisolvent for each polymer component, the choice of temperature for dissolution and precipitation, and the selection of an optimal solvent/antisolvent ratio for precipitation. These considerations impact process design and corresponding process economics. The key information needed to quantify tradeoffs between different operating conditions and solvent systems is the temperature-dependent solubility of each polymer in a proposed solvent system. Experimental data on the solubility of common polymers in some solvents are readily available. For example, Bloch summarized solvents and non-solvents for a large library of polymers,<sup>[22]</sup> while Zhao et al. reviewed solvent-based methods for waste plastics recycling<sup>[23]</sup> and listed solvents and nonsolvents for PS, polycarbonate (PC), PE, PP, PET, acrylonitrile butadiene styrene (ABS), and PVC with detailed experimental conditions. However, experimental collection of solubility data in the large number of possible solvents is prohibitively time-consuming, limiting the design of solvent systems for selective dissolution.

To more rapidly screen potential solvent systems, various empirical solvent parameters are available to identify possible solvents/nonsolvents for a given polymer; examples include Hildebrand,<sup>[24]</sup> Hansen,<sup>[25]</sup> Kamlet-Taft,<sup>[26]</sup> Gutmann,<sup>[27]</sup> and Swain<sup>[28]</sup> parameters. Among these, Hansen solubility parameters (HSPs) are the most widely used for selecting solvents for polymer dissolution. Tabulated HSP values are available for most common polymers and solvents, and solvent screening based on HSPs has many successful applications.<sup>[29]</sup> Computational methods can also be utilized for solvent selection. For example, Chandrasekaran et al. trained a deep neural network for the binary classification of solvents/nonsolvents using over

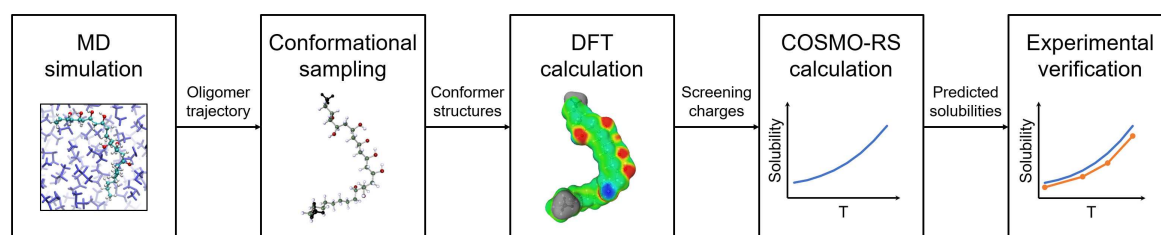
4500 homopolymers and 24 common solvents with an accuracy of 93.8%.<sup>[30]</sup> In general, these various approaches focus on qualitatively distinguishing solvents and nonsolvents for a given polymer at room temperature and are most accurate for single-component solvent systems. Quantitatively predicting temperature-dependent solubilities remains a challenge, which is the gap this research aims to fill.

Herein, we demonstrate a computational approach to quantitatively predict temperature-dependent polymer solubilities in different solvent systems. This approach improves upon the methodology that we previously employed to guide solvent design for the STRAP process,<sup>[8]</sup> which was suitable for qualitatively identifying potential solvent systems but overestimated polymer solubilities. Our approach employs classical molecular dynamics (MD) simulations and the conductor-like screening model for real solvents (COSMO-RS). We first model a polymer as a single oligomer molecule and simulate its motion in a dilute solution by MD simulation to obtain various conformations of the oligomer molecule. From this trajectory, we select representative conformations that span a range of structural parameters and use these as input to COSMO-RS for solubility calculations. We show that systematic variation of numerous simulation parameters leads to solubility predictions in good agreement with experimental solubility measurements for PE and EVOH in DMSO, water, tetrahydrofuran (THF), hexane, ethyl acetate, ethanol, acetone, toluene, and dimethylformamide (DMF). These predictions also outperform qualitative solubility predictions obtained from analysis with HSPs. We then demonstrate the applicability of the approach by screening a large library of 524 solvents at different temperatures to identify selective solvents for PE and EVOH dissolution. This computational approach thus can be used to screen solvents for solvent-based polymer recycling schemes and can be performed rapidly enough to guide process design.

## Experimental Section

### Summary of computational approach

A series of computational methods were used to evaluate polymer solubilities and compare them to experimental measurements for validation. Figure 1 summarizes this process in 5 steps:



**Figure 1.** Flow diagram summarizing the methods for computational polymer solubility prediction and experimental validation.

1. A MD simulation of a single oligomer was performed in dilute solution to obtain a simulation trajectory of various oligomer conformations.
2. A set of conformers was selected from the MD trajectory to span a range of representative oligomer structures.
3. Density functional theory (DFT) calculations were performed for the selected conformers to obtain corresponding screening charge density profiles.
4. Screening charge density profiles were input to COSMO-RS solubility calculations.
5. Experimental solubility measurements were used to benchmark the computational predictions.

This procedure was performed for two polymers: a homopolymer (PE) and copolymer (EVOH). Each step is detailed in the sections below.

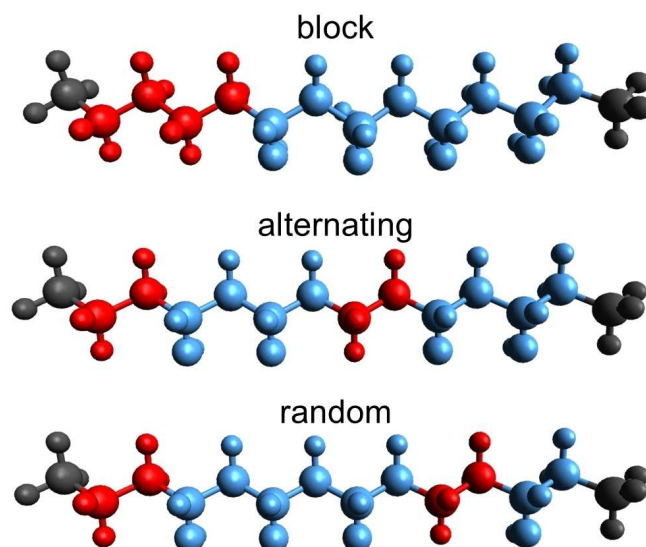
### Molecular dynamics simulations

Atomistic MD simulations were performed in the isothermal-isobaric ensemble using *Gromacs 2016*.<sup>[31]</sup> Each MD simulation contained 1 oligomer molecule (PE or EVOH) and 216 solvent molecules. PE was simulated in toluene and EVOH in water. All polymer and solvent molecules were parameterized using Antechamber and the Generalized AMBER force fields.<sup>[32,33]</sup> The simulation system was initialized with a cubic box containing a single polymer. The system was then solvated, energy minimized, and equilibrated for 2 ns in an *NPT* simulation at 300 K and 1 bar using a velocity-rescale thermostat and Berendsen barostat. A 10 ns *NPT* simulation was then performed at the same temperature and pressure using the Nose-Hoover thermostat and Parrinello-Rahman barostat. All simulations were performed using a leapfrog integrator with a 2-fs timestep. Verlet lists were generated using a 1.2 nm neighbor list cutoff. Van der Waals interactions were modeled with a shifted Lennard-Jones potential and Verlet cutoff-scheme that was smoothly shifted to zero at 1.2 nm. Electrostatic interactions were calculated using the smooth Particle Mesh Ewald method with a short-range cutoff of 1.2 nm, grid spacing of 0.14 nm, and 4th order interpolation. Bonds were constrained using the LINCS algorithm. All thermostats used a 2.0 ps time constant and all barostats used a 2.0 ps time constant with an isothermal compressibility of  $3.0 \times 10^{-5} \text{ bar}^{-1}$ .

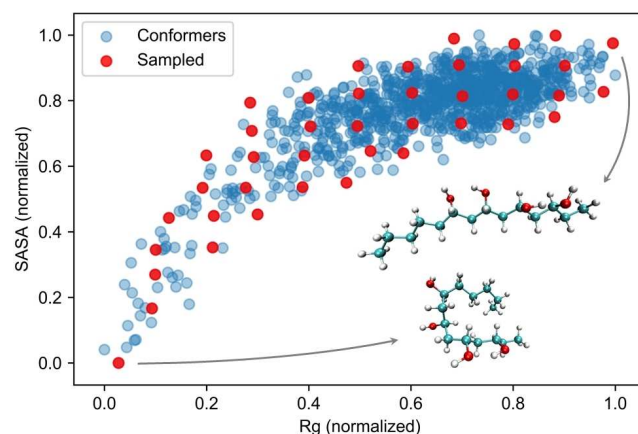
The chain lengths of the PE oligomers studied in this work range from 4 to 24 carbon atoms, whereas the chain lengths of EVOH range from 8 to 32 carbon atoms. EVOH is a copolymer that has two different monomer units (ethylene and vinyl alcohol). The EVOH studied experimentally (described below) has a molar ratio of 35%:65% ethylene/vinyl alcohol. Therefore, we modeled EVOH oligomers using a 1:2 ethylene/vinyl alcohol ratio and tested 3 different structures (block, alternating, and random) as shown in Figure 2.

### Conformational sampling and conformer selection

The MD trajectory of the oligomer molecule samples a large number of conformations as a function of simulation time. Since it is prohibitive to input all these conformers to COSMO-RS, we selected a set of representative conformers based on two structural parameters: the radius of gyration ( $R_g$ ) and the solvent-accessible surface area (SASA). We hypothesized that sampling conformers that capture a range of these parameters would provide reliable data to COSMO-RS for solubility predictions (see below). Figure 3 shows values in the two-dimensional  $R_g$ -SASA space generated from the MD trajectory of an EVOH oligomer.  $R_g$  and SASA values



**Figure 2.** Three types of structures of EVOH oligomer with 2 ethylene units (red) and 4 vinyl alcohol units (blue). Oligomers are modeled to have end methyl groups (black) that are deactivated such that they do not contribute to the COSMO-RS calculations.



**Figure 3.**  $R_g$ -SASA plot of an EVOH oligomer in MD simulation. Each blue dot represents a conformer, red dots are selected samples. The structures of two extreme conformations are shown: the folded one represents a bottom left dot, the elongated one represents a top right dot.

are normalized to a range from 0 to 1. Each blue dot corresponds to a conformer. We selected a set of conformers from this space by superimposing a square grid over the point cloud and selecting conformers closest to each grid intersection; sampled points are shown in red in Figure 3. The grid density can be tuned to control the number of conformer samples. Alternative methods for sampling conformers are discussed in the Supporting Information. In this work, 30 conformers of PE and 40 conformers of EVOH were studied.

### DFT calculations

Each conformer chosen from the MD trajectory was input to a DFT calculation to generate screening charge densities (COSMO files) using *Gaussian 16*.<sup>[34]</sup> Geometry optimization was first performed in implicit water using the conductor-like polarizable continuum

model (CPCM) at the BVP86/TZVP/DGA1 level of theory following past work.<sup>[35–37]</sup> A single-point calculation with the same DFT level was then performed to obtain screening charges in the infinite dielectric constant limit. COSMO files for the 9 solvents (DMSO, water, THF, hexane, ethyl acetate, ethanol, acetone, toluene, and DMF) studied experimentally in this work were also obtained using the same DFT calculations. For the 524 solvents involved in the large-scale solvent screening application, we used pre-calculated COSMO files from the database COSMObase-1901-BP-TZVP.

### Solubility prediction with COSMO-RS

COSMO files from the DFT calculations were input to COSMO-RS for property evaluation. COSMO-RS predicts the thermodynamic properties of multicomponent systems based on quantum mechanical calculations and statistical thermodynamics methods.<sup>[38,39]</sup> The chemical properties of each molecule are represented by the probability distribution of the screening charge densities (called the  $\sigma$ -profile) which arise at the molecular surface due to polarization within a virtual conductor environment.  $\sigma$ -profiles are obtained using DFT calculations as described in the preceding section. Once  $\sigma$ -profiles are known for all system components, the chemical potential of the solute is calculated based on molecular surface contacts and used to determine solubilities, solvent boiling points, and other thermodynamic properties. COSMO-RS was originally developed for liquid-liquid and liquid-vapor systems, and has been extended to solid compounds<sup>[40,41]</sup> and polymers.<sup>[42]</sup> We used  $\sigma$ -profiles of oligomer molecules with deactivated terminal groups to approximate the  $\sigma$ -profile of a polymer.<sup>[43]</sup> Since the screening charge density and resulting  $\sigma$ -profile depend on the specific molecular conformation, the MD simulation and conformational sampling steps described above were used to generate a set of oligomer conformations that span a range of possible  $\sigma$ -profiles.  $\sigma$ -Profiles for all selected conformers were then used to calculate the chemical potential of the polymer to enable predictions of polymer solubility via a solid-liquid equilibrium calculation [Eq. (1)]:

$$x_j = e^{(\mu_j^{\text{pure}} - \mu_j^{\text{solvent}} - \Delta G_{\text{fus}})/RT} \quad (1)$$

$\mu_j^{\text{pure}}$  is the chemical potential of pure compound  $j$  and  $\mu_j^{\text{solvent}}$  is the chemical potential of compound  $j$  in solvent at infinite dilution.  $\Delta G_{\text{fus}}$  is the free energy of fusion for compound  $j$  and accounts for the free-energy difference between the subcooled liquid state and ordered solid state, which can be obtained from either the heat of fusion or an experimentally measured reference solubility. Additional details on this calculation are discussed in the Supporting Information. All COSMO-RS calculations were performed using the COSMOtherm 19 software with the BP\_TZVP\_19 parameterization.<sup>[40,41,44]</sup>

### Experimental measurements of solubility

Experimental solubility measurements for PE and EVOH in DMSO, water, THF, hexane, ethyl acetate, ethanol, acetone, toluene, and DMF were performed to benchmark the computational predictions. The resins are LDPE and EVOH35 (molar ratio 35%:65% ethylene/vinyl alcohol). These solvents were selected because they are commonly used in experiments, while the polymers are common components of flexible plastic packaging. The experimental set up consisted of a 250 mL round bottom flask connected to a reflux condenser with a cold-water supply line. The round bottom flask contained a specific solvent amount for the experiment ( $\approx 30$  g). In each experiment, the round bottom flask was partially submerged in a 1500 mL dish containing silicone oil as a heat transfer fluid. The system was heated to the desired polymer dissolution temperature

with an electric heat plate equipped with a magnetic stir drive and the stirring rate was adjusted to have constant mixing. After the solvent reached the dissolution temperature, 1 wt% of polymer virgin pellets (with respect to the solvent) was added to the round bottom flask and mixed at a constant rate. If the polymer or part of the polymer was dissolved after 30 min, an additional 1 wt% of polymer pellets was added to the solvent. This procedure was repeated every 30 min. Once no further polymer could be dissolved, an additional 1 wt% was added to ensure an excess of undissolved polymer. After the final 30 min, the round bottom flask was removed from the oil. The liquid contents were poured into an antisolvent, either acetone or water, to precipitate the dissolved polymer. The precipitated material was filtered out of the solvent mixture, washed with water, and vacuum dried overnight. The undissolved solids remaining in the flask were also washed with water and vacuum dried overnight. The fractions were weighed after drying and the solubility was determined based on the amount of dissolved polymer with respect to the amount of solvent used. In some cases, the solubilities were regarded as a lower limit since there was a viscosity increase of the solvent (such cases are indicated below). The mass of the undissolved solids was included to complete the mass balance in each experiment.

## Results and Discussion

### Experimental solubility data

To develop a new workflow for quantitative polymer solubility prediction, we first present experimental measurements of PE and EVOH solubility in nine common solvents to provide benchmarks for comparison to COSMO-RS and HSP predictions. Table 1 shows experimental measurements of polymer solubility at elevated temperatures (close to the boiling point of each solvent).

### HSP-based solvent screening

We first utilized HSPs to qualitatively identify good and poor solvents for PE and EVOH to compare against the measured solubility data. The HSP system assigns each compound (polymer and solvent) three parameters that account for dispersion ( $\delta_D$ ), polar ( $\delta_P$ ), and hydrogen-bonding forces ( $\delta_H$ ). Solvent-polymer interactions are measured by their distance

**Table 1.** Experimental measurement of EVOH and PE solubilities.

| Solvent       | $T$<br>[°C] | PE solubility<br>[wt %] | EVOH solubility<br>[wt %] |
|---------------|-------------|-------------------------|---------------------------|
| DMSO          | 95          | 0.040                   | 24.020                    |
| water         | 95          | 0.002                   | 0.000                     |
| THF           | 65          | 0.018                   | 0.019                     |
| hexane        | 65          | 0.031                   | 0.000                     |
| ethyl acetate | 77          | 0.004                   | 0.053                     |
| ethanol       | 78          | 0.011                   | 0.000                     |
| acetone       | 55          | 0.001                   | 0.001                     |
| toluene       | 110         | 14.56 <sup>[a]</sup>    | 0.000                     |
| DMF           | 100         | 0.009                   | 30.63 <sup>[a]</sup>      |

[a] These values represent the lower limits of polymer solubilities in the selected solvents because higher polymer concentrations resulted in a viscous solution that was difficult to stir.



( $R_a$ ) in the HSP space ( $2\delta_D-\delta_P-\delta_H$  space). Each polymer has an additional radius parameter,  $R_0$ , that defines a sphere in the HSP space. Solvents within this sphere ( $R_a/R_0 < 1$ ) are expected to dissolve the polymer, whereas solvents outside of this sphere ( $R_a/R_0 > 1$ ) are not expected to dissolve the polymer. HSPs for a wide range of polymers and solvents have been tabulated in the HSP handbook.<sup>[29]</sup> However, there are multiple sets of HSP values for PE. Table 2 shows seven sets of room-temperature HSP values that are listed for PE in the handbook as well as values for EVOH. The different HSP values for PE resins reflect differences in properties such as crystallinities and molecular weight distributions. Unfortunately, little information is provided in this literature regarding the most appropriate set of parameters.

Table 3 shows  $R_a/R_0$  values for the nine solvents tested experimentally for PE and for EVOH. Each value is formatted based on comparison to the experimental solubility data: values of  $R_a/R_0 < 1$  are bold if the experimental solubility is greater than 1 wt% and normal if the experimental solubility is less than 1 wt%, whereas values of  $R_a/R_0 > 1$  are bold if the experimental solubility is less than 1 wt% and normal if the experimental solubility is greater than 1 wt%. Excluding DMSO, the HSP values for EVOH generally were good predictors of experimental solubility data. The room-temperature DMSO value is also near the threshold and would likely be identified as a good solvent if temperature effects were included. However, different sets of HSPs for PE lead to different solvent screening results. Surprisingly, the HSPs of high-density (HD)PE resulted in a perfect match with the experiments, even though

the polymer used experimentally was actually LDPE. The data for PE (1) and LDPE (2) correctly classify 8/9 solvents, but PE (2), LDPE (1), and LDPE (3) perform worse with only 5–6 correct classifications each. Moreover, the errors are generally due to HSP values identifying poor solvents as good (i.e.,  $R_a/R_0$  values are erroneously low), which cannot be attributed to temperature effects. Since it can be unclear which of these values to select in advance, there are clear deficiencies with using HSP values for screening solvents for these polymers. Similarly, the solubilities of EVOH in DMSO and DMF are substantially higher than that of PE in toluene even though the ratio  $R_a/R_0$  for PE in toluene tends to be much lower, highlighting that HSPs cannot provide quantitative information on solubility predictions. These observations motivate the need for alternative methods for solubility predictions.

### PE solubility predictions

We sought to develop a quantitative solubility prediction workflow using COSMO-RS to improve upon the deficiencies of HSPs noted above. We first systematically compared the impact of several parameters by predicting solubilities for PE as a simple homopolymer for which HSPs were shown to be inaccurate. Since the computational modeling of a full-length polymer molecule is formidable, the polymers were modeled as oligomers. The oligomer chain length was thus the first parameter we considered: increasing the chain length will increase the computational cost of the DFT calculations (the most time-consuming step in the COSMO-RS workflow) but could increase prediction accuracy. Consequently, there is a potential tradeoff between prediction accuracy and computational time that must be quantified to guide the solubility prediction workflow.

To select a reasonable chain length, PE oligomers with 4, 8, 12, 16, 20, and 24 carbon atoms were selected. MD simulations were performed of each oligomer and 3 conformers of each length were randomly selected. Each conformer was then input separately to DFT and COSMO-RS solubility calculations, with the latter using literature values for the melting point and heat of fusion of PE.<sup>[45]</sup> Prediction accuracy was quantified by calculating the mean prediction error relative to the experimen-

**Table 2.** HSP values for PE and EVOH (multiple versions for PE were found).<sup>[a]</sup>

| Polymer   | $\delta_D$ | $\delta_P$ | $\delta_H$ | $R_0$ |
|-----------|------------|------------|------------|-------|
| PE (1)    | 16.0       | 0.8        | 2.8        | 3.2   |
| PE (2)    | 16.5       | 2.7        | 6.1        | 7.9   |
| LDPE (1)  | 16.3       | 5.9        | 4.1        | 8.2   |
| LDPE (2)  | 16.5       | 4.5        | 0.5        | 6.0   |
| LDPE (3)  | 15.3       | 5.3        | 2.5        | 10.1  |
| HDPE      | 18.0       | 0.0        | 2.0        | 2.0   |
| HDPE/LDPE | 17.5       | 4.3        | 8.3        | 3.9   |
| EVOH      | 20.5       | 10.5       | 12.3       | 7.3   |

[a] Data source: C. M. Hansen, *Hansen Solubility Parameters: A User's Handbook*. CRC Press, 2007.

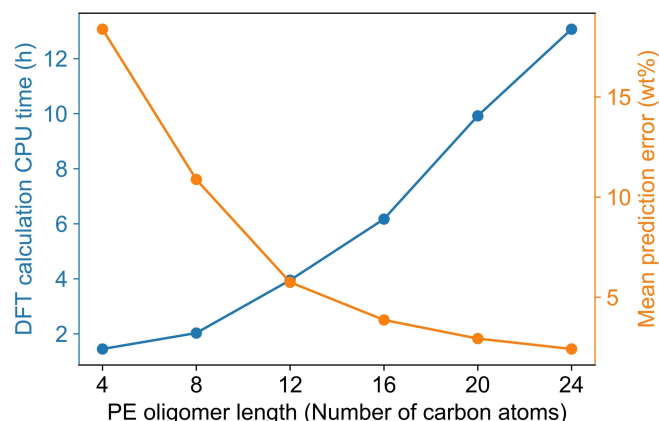
**Table 3.** HSP-based solvent screening for PE and EVOH.<sup>[a]</sup>

| Solvent       | HSP        |            |            | Solvent-polymer $R_a/R_0$ |             |             |             |             |              |             |             |
|---------------|------------|------------|------------|---------------------------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|
|               | $\delta_D$ | $\delta_P$ | $\delta_H$ | PE (1)                    | PE (2)      | LDPE (1)    | LDPE (2)    | LDPE (3)    | HDPE         | HDPE/LDPE   | EVOH        |
| DMSO          | 18.4       | 16.4       | 10.2       | <b>5.60</b>               | <b>1.87</b> | <b>1.57</b> | <b>2.64</b> | <b>1.47</b> | <b>9.18</b>  | <b>3.17</b> | 1.03        |
| water         | 15.5       | 16         | 42.3       | <b>13.23</b>              | <b>4.89</b> | <b>4.82</b> | <b>7.23</b> | <b>4.08</b> | <b>21.82</b> | <b>9.28</b> | <b>4.40</b> |
| THF           | 16.8       | 5.7        | 8.0        | <b>2.29</b>               | 0.46        | 0.49        | <b>1.27</b> | 0.62        | <b>4.31</b>  | 0.51        | <b>1.34</b> |
| hexane        | 14.9       | 0.0        | 0.0        | <b>1.14</b>               | 0.94        | 0.94        | 0.92        | 0.59        | <b>3.26</b>  | <b>2.74</b> | <b>2.69</b> |
| ethyl acetate | 15.8       | 5.3        | 7.2        | <b>1.97</b>               | 0.40        | 0.40        | <b>1.15</b> | 0.48        | <b>4.32</b>  | 0.95        | <b>1.63</b> |
| ethanol       | 15.8       | 8.8        | 19.4       | <b>5.76</b>               | <b>1.86</b> | <b>1.90</b> | <b>3.24</b> | <b>1.71</b> | <b>9.99</b>  | <b>3.19</b> | <b>1.63</b> |
| acetone       | 15.5       | 10.4       | 7.0        | <b>3.29</b>               | <b>1.01</b> | 0.68        | <b>1.50</b> | 0.67        | <b>6.29</b>  | <b>1.90</b> | <b>1.55</b> |
| toluene       | 18.0       | 1.4        | 2.0        | <b>1.29</b>               | <b>0.66</b> | <b>0.73</b> | <b>0.76</b> | <b>0.66</b> | <b>0.70</b>  | <b>1.80</b> | <b>2.00</b> |
| DMF           | 17.4       | 13.7       | 11.3       | <b>4.91</b>               | <b>1.56</b> | <b>1.32</b> | <b>2.38</b> | <b>1.27</b> | <b>8.30</b>  | <b>2.53</b> | <b>0.97</b> |

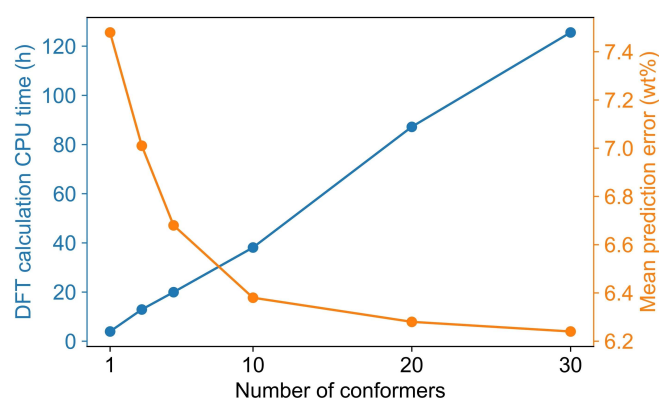
[a] Bold text indicates that the HSP prediction is consistent with the experimentally determined solubilities in Table 1, whereas normal text indicates that the HSP prediction is inconsistent with the experimentally determined solubilities.

tally measured solubilities in nine solvents listed in Table 1. The CPU time of the DFT calculation was used to quantify computational cost because the MD simulation time is not substantially affected by the chain length and the COSMO-RS calculation time is minimal. Figure 4 shows the mean prediction error and DFT calculation CPU time as a function of oligomer chain length. As the PE oligomer length increases, the mean solubility prediction error decreases and DFT CPU time increases, illustrating the hypothesized tradeoff between accuracy and computational cost. This result also confirms that the COSMO-RS prediction accuracy can be systematically improved. We selected the PE oligomer with 12 carbon atoms (corresponding to 5 ethylene units and 2 methyl end groups) for further workflow development because it has an acceptable prediction error with relatively short computational time.

With the oligomer chain length determined, we next studied the impact of the number of conformers on the tradeoff between prediction accuracy and computational cost. Different chain conformations have different screening charge density profiles that can influence the solubility prediction results, with



**Figure 4.** Variation in the solubility prediction error and computational cost with PE oligomer length. Three conformers are used for the solubility predictions for each data point.



**Figure 5.** Variation in the solubility prediction error and computational cost with the number of PE conformers used. All points represent an average over 10 subsets of conformers for a PE oligomer with a 12-carbon chain length.

substantial differences possible given conformational flexibility of the oligomers. From the MD trajectory, we selected a set of 30 representative conformations that span a range of  $R_g$  and SASA values for the 12-carbon PE oligomer using the grid sampling method. A series of subsets of different sizes (1, 3, 5, 10, and 20 conformers) were then generated by randomly selecting each number of conformers from the 30. 10 subsets were generated for each chosen subset size. Figure 5 shows the mean prediction error and DFT calculation CPU time as a function of the number of conformers (defined following the approach of Figure 4), with each point representing the average over all 10 subsets. The computational time required to generate a conformer set linearly scales with the set size. The mean prediction error decreases as more conformers are used and the error values tend to plateau as the number of conformers increases, again indicating that the COSMO-RS predictions can be systematically improved. Based on this comparison, the conformer number was selected to be 10 for the final solubility predictions.

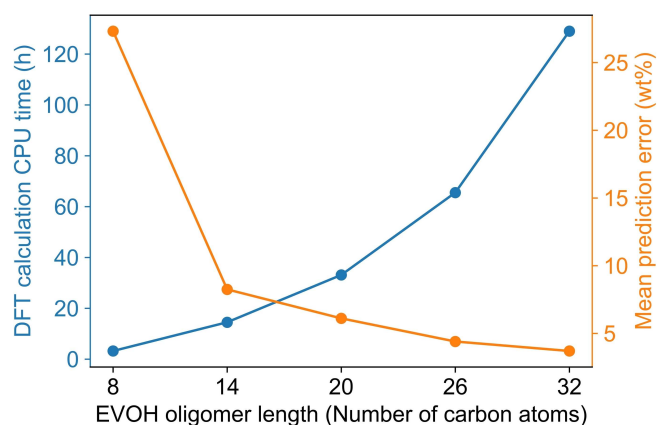
The 10-conformer set of the 12-carbon PE oligomer was used for final solubility predictions to compare to the experimental data. The solubility predictions in Figures 4 and 5 used the melting temperature and heat of fusion data of PE from literature sources, which leads to a relatively large error (Supporting Information).<sup>[45]</sup> However, the properties of PE resins could vary substantially between different samples, particularly since the heat of fusion will depend on polymer crystallinity. Therefore, we instead utilized the experimental solubility of PE in toluene at 110 °C as a reference value for the solid-liquid equilibrium calculation for all solvents. Table 4 shows corresponding predicted and experimental solubilities. As expected, the COSMO-RS protocol predicts near-zero solubilities for all solvents other than toluene in good agreement with the experimental measurements. Notably, several of these solvents are predicted by some sets of HSPs to be good solvents (e.g., hexane, THF, and ethyl acetate; see Table 3), indicating that the COSMO-RS approach qualitatively outperforms the HSP approach as well as providing quantitative solubility estimates. The COSMO-RS approach does slightly overestimate solubilities in nonsolvents, but these results are highly consistent with the experimental benchmarks if 2 wt% is used as a threshold to distinguish solvents and nonsolvents.

**Table 4.** COSMO-RS solubility predictions for PE compared to experimental measurements. Toluene is used as a reference and hence its prediction is exact.

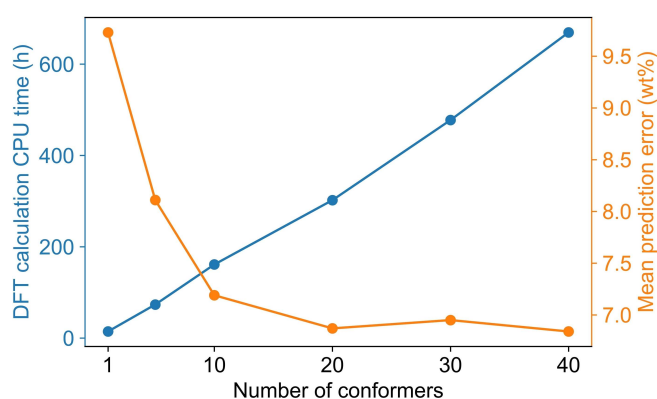
| Solvent       | <i>T</i><br>[°C] | Experimental solubility<br>[wt %] | COSMO-RS predictions<br>[wt %] |
|---------------|------------------|-----------------------------------|--------------------------------|
| DMSO          | 95               | 0.040                             | 0.34                           |
| water         | 95               | 0.002                             | 0.00                           |
| THF           | 65               | 0.018                             | 0.54                           |
| hexane        | 65               | 0.031                             | 0.85                           |
| ethyl acetate | 77               | 0.004                             | 0.52                           |
| ethanol       | 78               | 0.011                             | 0.20                           |
| acetone       | 55               | 0.001                             | 0.06                           |
| toluene       | 110              | 14.56                             | 14.56                          |
| DMF           | 100              | 0.009                             | 1.85                           |

## EVOH solubility predictions

We performed a similar process to compare the influence of oligomer length and the number of conformers on solubility predictions for EVOH. Since EVOH is a copolymer, we utilized the block copolymer oligomer structure for predictions when varying these parameters (to ensure a consistent monomer sequence), then compared predictions using the block, alternating and random copolymer structures (Figure 2). Figure 6 shows the mean prediction error and computational cost as a function of oligomer chain lengths between 8 to 32 carbon atoms in increments of 6 carbon atoms. This increment corresponds to 3 monomer units to maintain the 1:2 ethylene/vinyl alcohol ratio. For each oligomer length there are 3 conformers randomly selected from the MD trajectories. Based on the tradeoff between prediction accuracy and computational cost shown in this figure, the 14-carbon EVOH oligomer was selected because of the relatively low error values in a short computational time. This oligomer has 2 ethylene units, 4 vinyl alcohol units, and 2 methyl end groups.



**Figure 6.** Variation in the solubility prediction error and computational cost with EVOH oligomer length. Three conformers in the block copolymer structure are used for the solubility predictions for each data point.

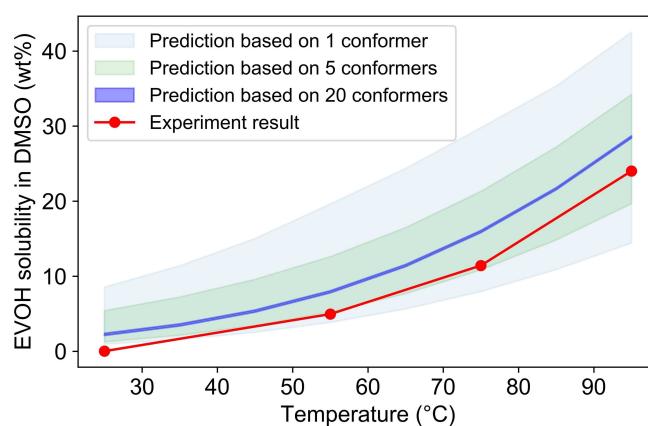


**Figure 7.** Variation in the solubility prediction error and computational cost with the number of EVOH conformers used. All points represent an average over 10 subsets of conformers for a EVOH oligomer with a 14-carbon chain length and block copolymer structure.

The number of conformers was then investigated for EVOH. A set of 40 representative conformations with various  $R_g$  and SASA values were selected using the grid sampling method and split into subsets of 1, 5, 10, 20, and 30 conformers. 10 distinct subsets were generated for each number of conformers. Figure 7 shows the mean prediction error and computational cost as a function of the number of conformers with each point representing the average over all 10 subsets. The scale of the vertical axis for mean prediction error is larger than that in Figure 5, indicating that the effect of the number of conformers is more significant for EVOH than for PE, but still plateaus as expected. Based on this comparison, the conformer number for EVOH was selected to be 20.

To further illustrate the effect of conformers on solubility predictions, Figure 8 shows the predicted EVOH solubility in DMSO as a function of temperature. Experimental data at four temperatures are included for reference. Predictions were based on either 1, 5, or 20 EVOH conformers, with 10 separate subsets of conformers used for predictions. Predicted solubilities with only a single conformer varied substantially and lie with the region indicated by the light blue color. The region bounding the solubility predictions decreased substantially for the 5-conformer sets (light green region). For the 20-conformer sets, the solubility calculations are highly consistent leading to a deep blue region that nearly converges to a solid line such that variations are difficult to discern by eye. These results indicate that increasing the number of conformers both leads to a decrease in prediction accuracy (Figure 7) and decreases the variation between different subsets of conformers, suggesting more robust results. Moreover, all conformer sets exhibit reasonable agreement with experiments, with the final results (for 20 conformers) slightly overestimating the measured solubilities but exhibiting similar trends.

Unlike PE, EVOH is a copolymer that has two different monomer units. The arrangement of these units leads to



**Figure 8.** Temperature-dependent solubility prediction for EVOH in DMSO with different conformer sets. The light blue, light green, and deep blue regions show the range of predicted solubilities obtained from 10 predictions using subsets of 1, 5, or 20 conformers, respectively. Red dots are experimental data. All predictions used an EVOH oligomer with a 14-carbon chain length and block copolymer structure. Results for each set of conformers are shown separately in Figure S4.

different types of structures. As shown in Figure 2, three types of structures were compared using oligomers with 14-carbon chain lengths and 20 conformers per oligomer. For this comparison, the experimental solubility of EVOH in DMSO at 95 °C was used as the reference for the solid-liquid equilibrium calculation. Table 5 compares solubility predictions to the experimental data. Clearly, DMF was recognized as a good solvent in all three cases, but only the block structure result correctly identifies it as the best solvent in the list. Solubilities are somewhat overpredicted for experimentally identified non-solvents, with ethanol being the most notable deviation. However, the values clearly differentiate solvents and non-solvents, with a slightly higher threshold (3 wt%) required to distinguish these categories than is necessary for PE. With respect to quantitative predictions, the random structure has the lowest overall mean prediction error, which is consistent with the fact that EVOH is a random copolymer.<sup>[46]</sup>

### Computational screening of selective solvents

To demonstrate the power of the proposed computational solubility prediction method, we applied the COSMO-RS protocol to identify selective solvents for PE and EVOH from a large library of 524 solvents. We note that the results in Tables 4 and 5 (and corresponding experiments) indicate that DMSO and DMF are selective solvents for EVOH (by promoting high

EVOH solubility and low PE solubility) whereas toluene is a selective solvent for PE; the screening procedure thus seeks to identify additional selective solvents beyond these examples. As above, a 10-conformer set of 12-carbon PE oligomers and a 20-conformer set of 14-carbon EVOH oligomers were used. The solubilities were computed at room temperature and 1 °C below the boiling points of the respective solvents. For this screening procedure, we define selective solvents as solvents with a solubility greater than 10 wt% for one polymer and lower than 1 wt% for the other. Table 6 shows some selected examples of identified selective solvents; full solubility prediction results are available in the Supporting Information. At room temperature, the PE solubility was < 1 wt% in all solvents considered, so no PE-selective solvents were found in this condition. PE-selective solvents were identified when temperatures were increased near the solvents' boiling points. EVOH-selective solvents were found at both room temperature and elevated temperatures. The calculations for this large-scale solvent screening were completed within 2 h since no new MD or DFT calculations were required, demonstrating the efficiency of this computational approach.

The computational efficiency of this solvent screening approach enables its application to the selection of solvents for solvent-mediated plastic recycling processes, such as the STRAP process described in our previous work. In our first STRAP process, pure DMSO was used to dissolve EVOH from a multilayer plastic film, and then water was added as an

**Table 5.** COSMO-RS solubility predictions for EVOH compared to experimental measurements. Three different copolymer structures are compared (Figure 2).

| Solvent       | <i>T</i><br>[°C] | Experimental solubility<br>[wt %] | COSMO-RS solubility predictions<br>Block<br>[wt %] | Alternating<br>[wt %] | Random<br>[wt %] |
|---------------|------------------|-----------------------------------|----------------------------------------------------|-----------------------|------------------|
| DMSO          | 95               | 24.02 <sup>[a]</sup>              | 24.02                                              | 24.02                 | 24.02            |
| water         | 95               | 0.00                              | 0.01                                               | 0.03                  | 0.03             |
| THF           | 65               | 0.02                              | 5.53                                               | 2.06                  | 1.78             |
| hexane        | 65               | 0.00                              | 0.02                                               | 0.00                  | 0.00             |
| ethyl acetate | 77               | 0.05                              | 2.47                                               | 0.66                  | 0.49             |
| ethanol       | 78               | 0.00                              | 7.63                                               | 3.94                  | 2.98             |
| acetone       | 55               | 0.00                              | 1.81                                               | 0.58                  | 0.45             |
| toluene       | 110              | 0.00                              | 4.49                                               | 0.64                  | 0.39             |
| DMF           | 100              | 30.63                             | 27.49                                              | 21.19                 | 20.48            |

[a] Experimental data in this solvent was used as reference for the solid-liquid equilibrium calculation, so all three structures have the same calculated solubility.

**Table 6.** COSMO-RS solvent screening for PE and EVOH.<sup>[a]</sup>

| Solvent              | <i>T</i><br>[°C] | PE solubility<br>[wt %] | EVOH solubility<br>[wt %] | Comment                            |
|----------------------|------------------|-------------------------|---------------------------|------------------------------------|
| hydrazine            | 25               | 0.00                    | 26.38                     | RT, EVOH-selective                 |
| ethylenediamine      | 25               | 0.00                    | 24.40                     |                                    |
| <i>n</i> -butylamine | 25               | 0.00                    | 15.26                     |                                    |
| octane               | 124.6            | 45.40                   | 0.13                      | near boiling point, PE-selective   |
| 1-octene             | 120.2            | 35.22                   | 0.21                      |                                    |
| tetrachloroethylene  | 120.3            | 21.34                   | 0.08                      | near boiling point, EVOH-selective |
| methylhydrazine      | 86.5             | 0.23                    | 34.21                     |                                    |
| acetic acid          | 116.9            | 0.62                    | 25.46                     |                                    |
| formic acid          | 100              | 0.02                    | 24.42                     |                                    |

[a] 524 solvents at two temperatures (RT and near each solvent's boiling point) were considered. Nine selective solvents are presented as examples (no PE-selective solvents were identified at room temperature). Complete results are presented in the Supporting Information.



antisolvent to precipitate the polymer.<sup>[8]</sup> The computational approach presented in this work is more accurate than the prior approach due to the newly developed methods for conformational screening and oligomer length selection, suggesting that it can be applied to improve the recovery of polymers in revised STRAP processes. For example, our prior work selected solvents for STRAP based primarily on solubility considerations; techno-economic analysis was then performed to evaluate the minimum selling price of the recovered materials necessary for economic feasibility. The computational screening approach could instead be applied to identify solvents that both selectively dissolve target polymers and reduce downstream costs associated with polymer recovery and solvent recycling, leveraging the computational efficiency and improved accuracy described in this work. The computational approach could also be utilized to identify temperature-controlled polymer dissolution and precipitation at fixed solvent compositions to reduce the use of antisolvent and improve process economics.<sup>[21]</sup> Another example application would be to search for green solvents (quantified by metrics such as energy demand and greenhouse gas emissions in solvent production or environmental health and safety rankings<sup>[47,48]</sup>) as potential replacements for common industrial solvents, permitting quantification of tradeoffs between greenness and solubility behavior of the solvents. There are thus multiple opportunities to apply the computational protocol developed in this work to efficiently identify solvent systems that improve selective dissolution processes for plastic recycling.

## Conclusions

In this work, we present a computational approach to quantitatively predict temperature-dependent polymer solubilities in various solvents. Our approach employs molecular dynamics (MD) simulations and conductor-like screening model for real solvents (COSMO-RS) calculations. The MD simulation models a polymer as an oligomer and simulates its motion in a dilute solution to generate various chain conformations. Representative oligomer conformers that span a range of radius of gyration ( $R_g$ ) and solvent-accessible surface area (SASA) values are then used as input structures to COSMO-RS solubility calculations. We note that while this work uses COSMO-RS, we expect that alternative COSMO-based models (e.g., COSMO-SAC) may also be suitable for this approach.<sup>[49]</sup> We studied the effects of oligomer length, conformer numbers, and copolymer structures on the accuracy of solubility predictions. We show that increasing the oligomer length and number of conformers can systematically improve prediction accuracy, albeit with increased computational cost. By selecting a reasonable oligomer length and number of conformers, the proposed approach leads to solubility predictions in good agreement with experimental measurements for polyethylene (PE) and ethylene vinyl alcohol (EVOH) in DMSO, water, THF, hexane, ethyl acetate, ethanol, acetone, toluene and DMF. These predictions outperform qualitative Hansen solubility parameter (HSP)-based solubility predictions while also enabling predic-

tions of temperature-dependent behavior, and lead to more accurate predictions than the protocol employed in our past work.<sup>[8]</sup> We demonstrated the applicability and efficiency of the new computational approach by screening 524 solvents to identify selective solvents for the dissolution of PE and EVOH at two temperatures. Future work will apply this protocol to additional polymers, including both industrially relevant polymers and biopolymers (e.g., lignin) that must be solubilized prior to depolymerization. A current limitation of the method is the need for a single experimental solubility measurement in a good solvent as input to account for sample-specific parameter. While a suitable reference good solvent could be identified by initial qualitative screening using HSP values, future work will aim to reduce the amount of needed experimental input by incorporating explicit consideration of polymer molecular weight and crystallinity.

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## Conflict of Interest

The authors declare no conflict of interest.

**Keywords:** molecular simulation · polymer recycling · polymer solubility · selective dissolution · solvent screening

- [1] L. K. Massey, *Permeability properties of plastics and elastomers: a guide to packaging and barrier materials*, Cambridge University Press, 2003.
- [2] D. Lithner, Å. Larsson, G. Dave, *Sci. Total Environ.* **2011**, 409, 3309–3324.
- [3] D. Kim, S. W. Kim, *Korean J. Chem. Eng.* **2003**, 20, 776–782.
- [4] D. Feldman, *J. Polym. Environ.* **2001**, 9, 49–55.
- [5] K. K. Mokwena, J. Tang, M.-P. Laborie, *J. Food Eng.* **2011**, 105, 436–443.
- [6] D. Langhe, M. Ponting, *Manufacturing and novel applications of multi-layer polymer films*, William Andrew, 2016.
- [7] J. K. Kim, S. Thomas, P. Saha, *Multicomponent Polymeric Materials*, Vol. 223, Springer, 2016.
- [8] T. W. Walker, N. Frelka, Z. Shen, A. K. Chew, J. Banick, S. Grey, M. S. Kim, J. A. Dumesic, R. C. Van Lehn, G. W. Huber, *Sci. Adv.* **2020**, 6, eaba7599.
- [9] J. M. Garcia, M. L. Robertson, *Science* **2017**, 358, 870–872.
- [10] O. Horodytska, F. J. Valdés, A. Fullana, *Waste Manage.* **2018**, 77, 413–425.
- [11] P. Lacy, J. Rutqvist, *Waste to wealth: The circular economy advantage*, Springer, 2016.
- [12] R. Coles, M. J. Kirwan, *Food and beverage packaging technology*, Wiley, 2011.
- [13] U.S. Environmental Protection Agency, 'Containers and packaging: Product-specific data', can be found under [www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/containers-and-packaging-product-specific-data](http://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/containers-and-packaging-product-specific-data), 2021.
- [14] M. Crippa, B. De Wilde, R. Koopmans, J. Leyssens, J. Muncke, A. Ritschkoff, K. Van Doorselaer, C. Velis, M. Wagner, *A circular economy*

- for plastics: Insights from research and innovation to inform policy and funding decisions, 2019.
- [15] M. Pohjakallio, T. Vuorinen, A. Oasmaa in *Plastic Waste and Recycling*, Elsevier, 2020, pp. 359–384.
- [16] J. M. Hoffman, *Mach. Des.* **2003**, *75*, 79–79.
- [17] Plasteurope, 'VINYLLOOP: Closure of operation in Italy / Phthalates issue under REACH brings down European PVC recycling project', can be found under [https://www.plasteurope.com/news/Closure\\_of\\_operation\\_in\\_Italy\\_Phthalates\\_issue\\_under\\_REACH\\_brings\\_down\\_PVC\\_recycling\\_project/](https://www.plasteurope.com/news/Closure_of_operation_in_Italy_Phthalates_issue_under_REACH_brings_down_PVC_recycling_project/), 2018.
- [18] I. Vollmer, M. J. Jenks, M. C. Roelands, R. J. White, T. van Harmelen, P. de Wild, G. P. van Der Laan, F. Meirer, J. T. Keurentjes, B. M. Weckhuysen, *Angew. Chem. Int. Ed.* **2020**, *59*, 15402–15423; *Angew. Chem.* **2020**, *132*, 15524–15548.
- [19] M. Schlummer, F. Wolff, A. Mäurer in 2016 Electronics Goes Green 2016 + (EGG), IEEE, **2016**, pp. 1–6.
- [20] R. Côté, *Processes for recycling polystyrene waste*, **2017**, 20170298207.
- [21] K. L. Sánchez-Rivera, P. Zhou, M. S. Kim, L. D. González Chávez, S. Grey, K. Nelson, S.-C. Wang, I. Hermans, V. M. Zavala, R. C. Van Lehn, G. W. Huber, unpublished results, **2021**, 10.1002/cssc.202101128.
- [22] D. R. Bloch, *The Wiley Database of Polymer Properties* **2003**.
- [23] Y.-B. Zhao, X.-D. Lv, H.-G. Ni, *Chemosphere* **2018**, *209*, 707–720.
- [24] J. H. Hildebrand, *Proc. Natl. Acad. Sci. USA* **1950**, *36*, 7.
- [25] C. M. Hansen, L. Just, *Ind. Eng. Chem. Res.* **2001**, *40*, 21–25.
- [26] R. W. Taft, J.-L. M. Abboud, M. J. Kamlet, M. H. Abraham, *J. Solution Chem.* **1985**, *14*, 153–186.
- [27] V. Gutmann, *Electrochim. Acta* **1976**, *21*, 661–670.
- [28] C. G. Swain, M. S. Swain, A. L. Powell, S. Alunni, *J. Am. Chem. Soc.* **1983**, *105*, 502–513.
- [29] C. M. Hansen, Hansen solubility parameters: A user's handbook, CRC Press, **2007**.
- [30] A. Chandrasekaran, C. Kim, S. Venkatram, R. Ramprasad, *Macromolecules* **2020**, *53*, 4764–4769.
- [31] M. J. Abraham, T. Murtola, R. Schulz, S. Páll, J. C. Smith, B. Hess, E. Lindahl, *SoftwareX* **2015**, *1*, 19–25.
- [32] J. Wang, W. Wang, P. A. Kollman, D. A. Case, *J. Mol. Graphics Modell.* **2006**, *25*, 247–260.
- [33] J. Wang, R. M. Wolf, J. W. Caldwell, P. A. Kollman, D. A. Case, *J. Comput. Chem.* **2004**, *25*, 1157–1174.
- [34] Gaussian 16, M. Frisch, G. Trucks, H. Schlegel, G. Scuseria, M. Robb, J. Cheeseman, G. Scalmani, V. Barone, G. Petersson, H. Nakatsuji, *Gaussian Inc.* **2016**.
- [35] J. Palomar, V. R. Ferro, J. S. Torrecilla, F. Rodríguez, *Ind. Eng. Chem. Res.* **2007**, *46*, 6041–6048.
- [36] Y.-R. Liu, K. Thomsen, Y. Nie, S.-J. Zhang, A. S. Meyer, *Green Chem.* **2016**, *18*, 6246–6254.
- [37] X. Liu, Y. Nie, Y. Liu, S. Zhang, A. L. Skov, *ACS Sustainable Chem. Eng.* **2018**, *6*, 17314–17322.
- [38] A. Klamt, *J. Phys. Chem.* **1995**, *99*, 2224–2235.
- [39] A. Klamt, V. Jonas, T. Bürger, J. C. Lohrenz, *J. Phys. Chem. A* **1998**, *102*, 5074–5085.
- [40] A. Klamt, F. Eckert, M. Hornig, M. E. Beck, T. Bürger, *J. Comput. Chem.* **2002**, *23*, 275–281.
- [41] F. Eckert, A. Klamt, *AIChE J.* **2002**, *48*, 369–385.
- [42] C. Loschen, A. Klamt, *Ind. Eng. Chem. Res.* **2014**, *53*, 11478–11487.
- [43] J. Kahlen, K. Masuch, K. Leonhard, *Green Chem.* **2010**, *12*, 2172–2181.
- [44] COSMOlogic GmbH & Co KG, COSMOtherm Version 19.0.5 (Revision 4825–4826), **2019**.
- [45] B. Wunderlich, *Thermal Analysis*, Academic Press, San Diego, CA, **1990**.
- [46] C. Maes, W. Luyten, G. Herremans, R. Peeters, R. Carleer, M. Buntinx, *Polym. Rev.* **2018**, *58*, 209–246.
- [47] P. G. Jessop, *Green Chem.* **2011**, *13*, 1391–1398.
- [48] F. P. Byrne, S. Jin, G. Paggiola, T. H. M. Petchey, J. H. Clark, T. J. Farmer, A. J. Hunt, C. R. McElroy, J. Sherwood, *Sustainable Chem. Processes* **2016**, *4*, 1–24.
- [49] S.-T. Lin, S. I. Sandler, *Ind. Eng. Chem. Res.* **2002**, *41*, 899–913.

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